

Unsupervised Learning

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By the end of this session, you will be able to:

- ▶ Differentiate between Supervised and Unsupervised Learning paradigms.
- ▶ Understand K-Means clustering, and how it could be applied to discover hidden groups in data.
- ▶ Analyze high-dimensional data using PCA and t-SNE for visualization and compression.
- ▶ Understand how Autoencoders utilize neural networks for tasks like denoising, generation, and anomaly detection.

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What is Unsupervised Learning?

So far: We've been doing **Supervised Learning**, learning from labeled data (x, y) .

Supervised Learning

Data: (x, y) pairs

- ▶ x : input (features)
- ▶ y : label (target)

Goal: Learn $f(x) \approx y$

Examples:

- ▶ Image classification
- ▶ House price prediction

Unsupervised Learning

Data: Only x (no labels!)

- ▶ Just raw data
- ▶ No target to predict

Goal: Find hidden patterns or structure

Examples:

- ▶ Customer segmentation
- ▶ Data compression

Key difference: No labels $\rightarrow$ algorithm must discover structure on its own!

Labels are expensive!

- ▶ Labeling data requires human effort, time, and expertise
- ▶ Most real-world data is unlabeled
- ▶ Sometimes we don't even know what labels to look for

The Big Idea

Can we learn useful things from data **without** labels?

Yes! Unsupervised learning finds hidden patterns, structure, and representations.

What can we do without labels?

1. Clustering

Group similar data points together

Example: Customer segmentation, document group

3. Anomaly Detection

Identify unusual or outlier data points

Example: Fraud detection, quality control

2. Dimensionality Reduction

Compress high-dimensional data to lower dimensions

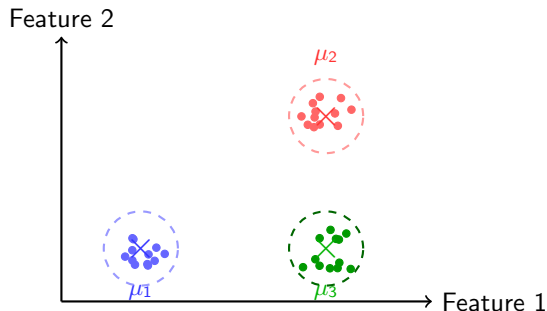
Example: Visualization, feature extraction

4. Generation

Learn to create new, realistic data samples

Example: Image synthesis, data augmentation

Main Idea: Represent each cluster by its center (mean)



- ▶ Each cluster has a **center** (centroid) μ_k (marked with $\times$)
- ▶ Points belong to their **nearest** center
- ▶ Goal: Find centers that minimize total distance to assigned points

K-Means Algorithm

K-Means Algorithm

Input: Data $\{x_1, x_2, \dots, x_n\}$ and number of clusters K

Initialize: Randomly place K cluster centers $\mu_1, \mu_2, \dots, \mu_K$

Repeat until convergence:

1. **Assignment Step:** For each point x_i , assign to nearest center

$$c_i = \arg \min_{k=1, \dots, K} \|x_i - \mu_k\|^2$$

(find which center is closest)

2. **Update Step:** Move each center to mean of assigned points

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} x_i$$

(average all points in cluster k)

Stop when: Assignments don't change OR max iterations reached

K-Means: Step-by-Step Example



Repeat steps 2-3 until centers stop moving (typically 10-100 iterations)

Problem: How many clusters should we use?

Elbow Method

Plot "within-cluster sum of squares" (WCSS) vs K

$$WCSS = \sum_{k=1}^K \sum_{i \in C_k} \|x_i - \mu_k\|^2$$



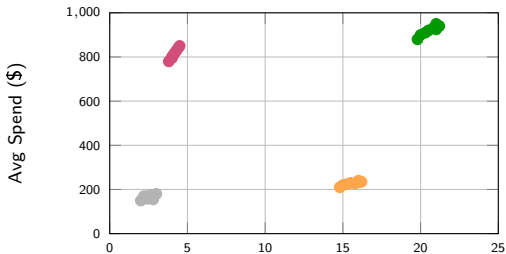
Look for the "elbow" where adding more clusters doesn't help much (in the above case it is 4)

Application 1: Customer Segmentation

Use case: Group customers for targeted marketing

Example: E-commerce Data

- Features: frequency, spend, recency
- Run K-Means with $K = 4$
- Interpret clusters



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Discovered Segments:

Inactive

Low freq,
low spend

Action:
Re-engage

Bargain

High freq,
low spend

Action:
Discounts

Big Spenders

Low freq,
high spend

Action:
Premium Service

VIP

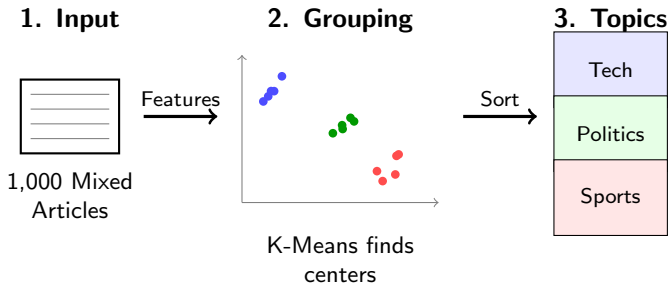
High freq,
high spend

Action:
Loyalty Program

Unsupervised Learning

Application 2: Document Clustering

Use case: Automatically organize news articles by topic.

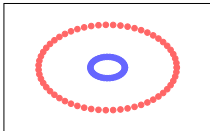


Process

- ▶ Convert text into numbers (Text Features).
- ▶ Similar stories appear close together in space.
- ▶ K-Means automatically groups them into topics.

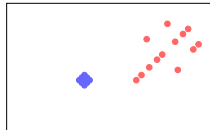
K-Means assumes spherical, equal-sized clusters

Problem 1: Non-spherical



Fails on concentric circles!

Problem 2: Different sizes



Struggles with size differences!

Other Limitations

- ▶ Sensitive to outliers
- ▶ Must specify K in advance
- ▶ Random initialization matters
- ▶ Bad results for high dimensional data

Alternatives: DBSCAN (density-based), Gaussian Mixture Models, Hierarchical clustering

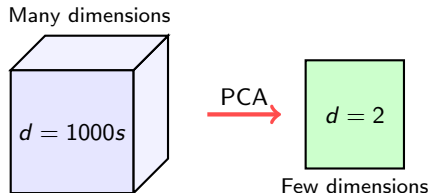
Real-world data is high-dimensional:

Examples:

- ▶ Images: $1000 \times 1000 = 1\text{M}$ pixels
- ▶ Text: 10,000+ words
- ▶ Genomics: 20,000+ genes
- ▶ Sensors: 100+ measurements

Problems:

- ▶ Hard to visualize
- ▶ Slow to compute
- ▶ Lots of redundancy
- ▶ Noisy measurements



Solution: Find a lower-dimensional representation while keeping important information (Dimensionality Reduction)

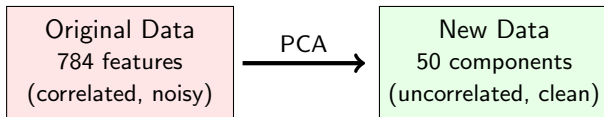
Principal Component Analysis (PCA) is a dimensionality reduction algorithm that finds the most important directions in your data.

Core Idea

Instead of using original features, create **new features** (principal components) that:

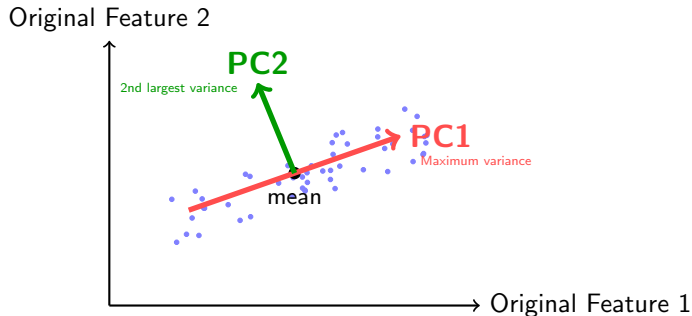
- ▶ Capture maximum variance
- ▶ Are uncorrelated with each other
- ▶ Are ranked by importance

Then keep only the top few components!



PCA Intuition: Finding the Best Directions

Goal: Find directions where data varies the most.



- ▶ PC1: Direction with most spread
- ▶ PC2: Perpendicular to PC1, next most spread
- ▶ PC3, PC4, ... : Each perpendicular to all previous ones

Step 1: Prepare the Data

1. Center the Data

Shift the data so the center is at the origin (0,0).

$$X_{\text{centered}} = X - \bar{X}$$

(Subtract the mean from every feature)

2. Compute Covariance Matrix

Calculate how features vary with respect to each other.

$$C = \frac{1}{n} X_{\text{centered}}^{\top} X_{\text{centered}}$$

(Finds correlations between features)

Step 2: Find Directions & Project

3. Find Principal Components

Calculate **Eigenvectors** and **Eigenvalues** of the covariance matrix C .

- ▶ **Eigenvectors:** The directions of the new axes.
- ▶ **Eigenvalues:** The amount of variance (information) along those axes.

4. Select & Project

Pick the top k eigenvectors with the largest eigenvalues and transform the data.

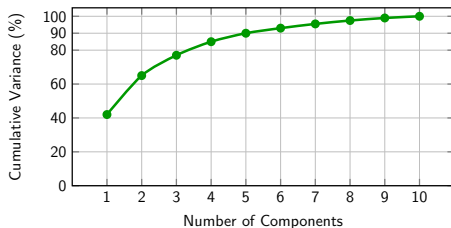
$$Z = X_{\text{centered}} \cdot V_k$$

(Reduces data from d dimensions $\rightarrow k$ dimensions)

Key question: How many components are enough?

Cumulative Variance Explained

(The most common method for selecting K)



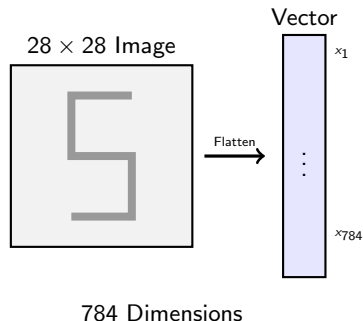
Rule of Thumb: Choose the smallest K that preserves **80–95%** of the total variance.

The Data: MNIST Digits, 70,000 handwritten digits (0–9).

The Challenge:

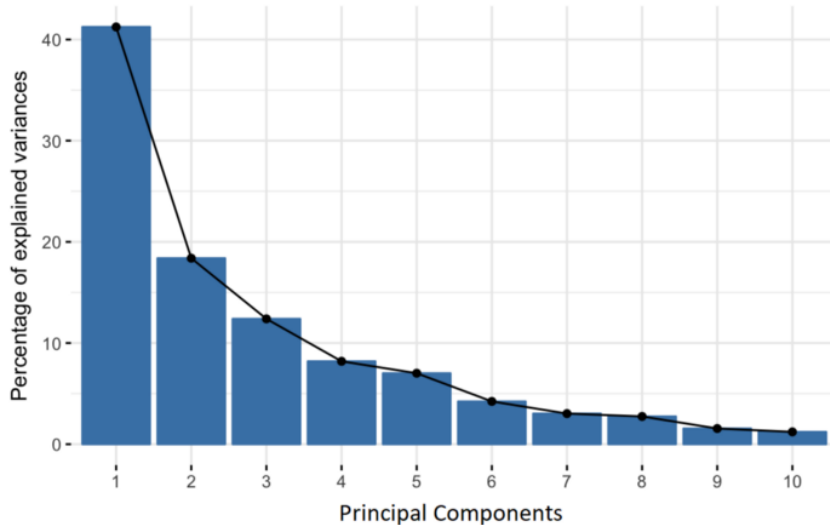
- ▶ Each image is 28×28 pixels.
- ▶ Total features: $28 \times 28 = \mathbf{784}$ dimensions!
- ▶ We cannot "see" in 784 dimensions.

Question: Can we compress this to just **2D** to see how the digits relate?



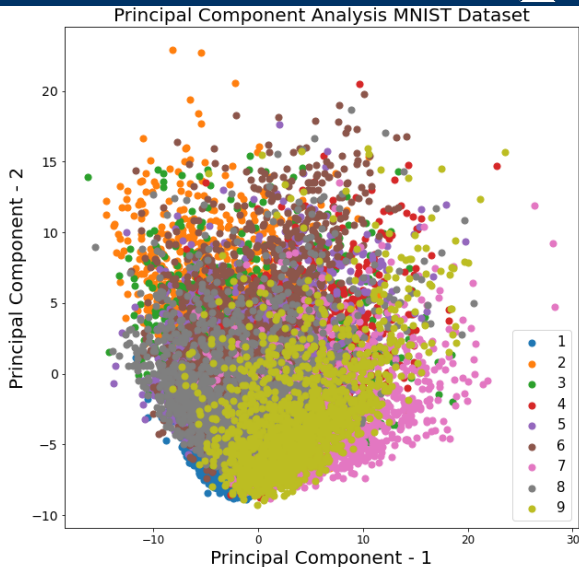
Applying PCA: We project the 784 dimensions down to the top 2 Principal Components.

Application 1: Data Visualization (cont.)

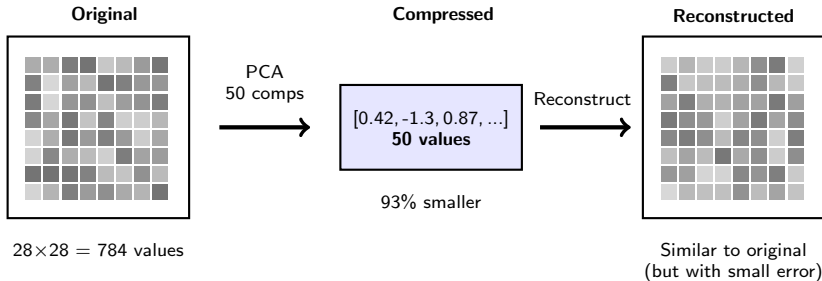


Is this the best visualization?

- ▶ It captures some structure, but points overlap heavily.
- ▶ **Reason:** PCA is **linear**, it cannot unfold complex, curved structures.
- ▶ PCA is mostly used for compression. We will cover a better visualization algorithm shortly.



Use case: Store data more efficiently



More components = better reconstruction, less compression

10 PCs
99% compression
(blurry)

50 PCs
93% compression
(good)

200 PCs
75% compression
(near perfect)

PCA Limitations:

- ▶ **Linear only:** Finds linear combinations of features
- ▶ **Sensitive to scale:** Must standardize data first
- ▶ **Not Interpretable:** PCs are combinations of all features
- ▶ **Outliers:** Can distort principal components

Let's have a look at a better algorithm for high dimensional data visualization...

PCA is great... but it's linear. Many datasets have **non-linear** structure.

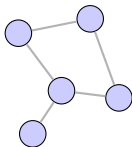
Problem

- ▶ High-dimensional data can be hard to visualize
- ▶ PCA may **mix** classes if the separation is non-linear
- ▶ We mainly want a **good 2D plot** for exploration

Main Idea

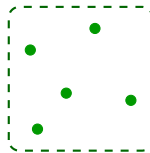
t-SNE tries to build a 2D map where: neighbors in high-D remain neighbors in 2D.

High-D



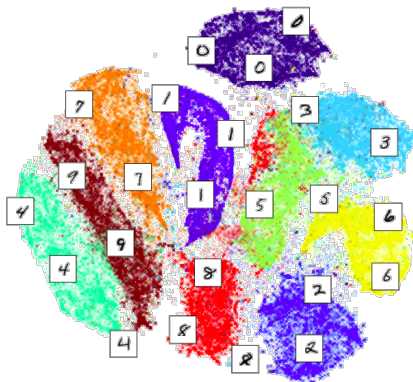
t-SNE
→

2D map



Neighborhood stays together

Example: Visualizing MNIST via t-SNE

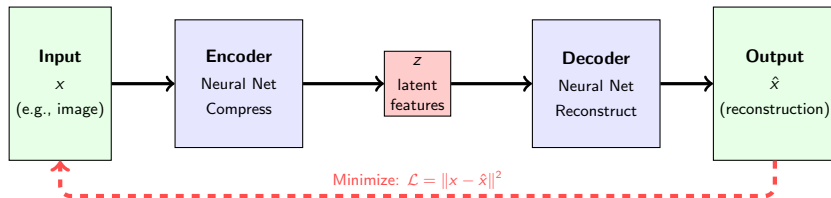


This is much better than PCA! We can see how similar digits are grouped together, while different ones are far from each other.

We have covered classical algorithms like K-Means and PCA.

Now, let's explore a powerful Neural Network architecture designed for unsupervised learning...

Autoencoder: a type of neural network that learn to compress data into a compact form and then reconstruct it to match the original input.



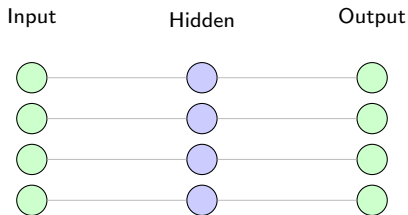
The trick: Force information through a narrow **bottleneck** z .

This forces the network to learn compressed, meaningful representations!

Why Does the Bottleneck Work?

What if there's no bottleneck?

Without Bottleneck

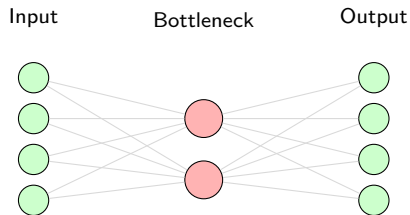


Network can just copy:

$$h_i = x_i, \hat{x}_i = h_i$$

Learns nothing useful!

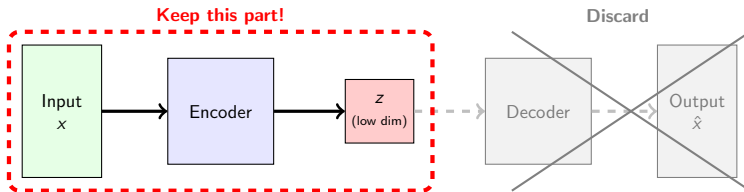
With Bottleneck



Must compress 4D $\rightarrow$ 2D
Forces learning of structure!

Learns meaningful features!

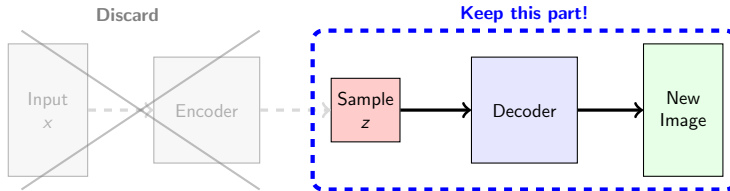
How to use an Autoencoder for compression:



The Process

1. **Train** the full network to reconstruct the input ($x \approx \hat{x}$).
2. **Discard** the Decoder (it's only needed for training).
3. **Use the Encoder** to map high-dimensional data x to low-dimensional code z .

How to use an Autoencoder for generation:

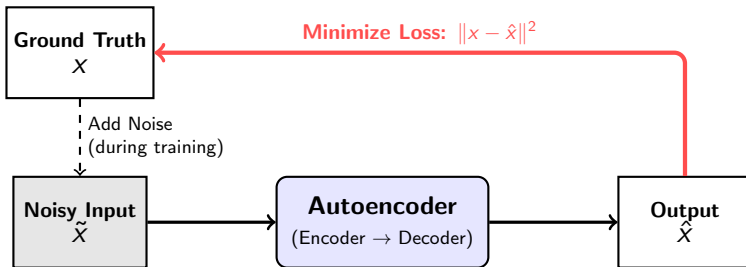


The Process

1. **Train** the full network on images to learn features.
2. **Discard** the Encoder.
3. **Sample** a random vector z (pick random numbers).
4. **Generate** a brand new image by running z through the Decoder.

Application 3: Denoising Autoencoders

Idea: Train the network to recover the original signal from corrupted input.

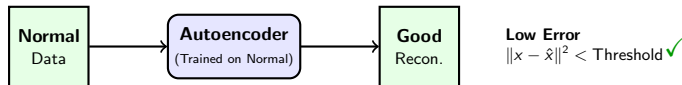


How it works

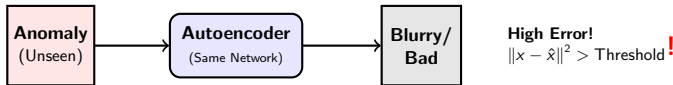
- ▶ Take clean image x .
- ▶ Corrupt it to get $\tilde{x}$.
- ▶ Force AE to map $\tilde{x} \rightarrow x$.

Concept: The network learns to reconstruct "Normal" data perfectly. When it sees an anomaly, it fails to reconstruct it.

1. Training / Normal Operation



2. Detection Phase



Real-World Use Cases

- ▶ **Credit Card Fraud:** Normal transactions reconstruct well, theft looks weird.
- ▶ **Manufacturing:** Detect defective parts on an assembly line.

- ▶ Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*
- ▶ Andrew Ng, *Machine Learning Specialization* (Coursera/DeepLearning.AI)

Slides contributed by Mohamed Eltayeb